Program : **Diploma in Chemical Engineering**	
Course Code : **5071**	Course Title: **Mass Transfer Operations II**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L: 3 T: 1 P: 0)**	Periods per semester: **60**

Course Objectives:

- To enable the students to apply the principles of distillation and extraction in process plant operations.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I Mathematics II	1 2
Knowledge of basic chemistry		Applied Chemistry Applied Chemistry II	1 2
Knowledge of unit operations and processes in chemical engineering		Chemical Process Principles	2
Knowledge of principles of mass transfer		Mass transfer operations I	4

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the principles of distillation operation.	13	Applying
CO2	Interpret the types of distillation.	20	Applying
CO3	Summarise leaching operation.	15	Understanding
CO4	Summarise liquid – liquid extraction.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Interpret the principles of distillation operation.**		
M1.01	Outline the principles of distillation	2	Understanding
M1.02	Illustrate vapour liquid diagram for a binary mixture.	2	Understanding
M1.03	Interpret relative volatility.	4	Applying
M1.04	Summarize azeotropes.	2	Understanding
M1.05	Summarize types of distillation.	3	Understanding
Contents: Distillation - Principle - Applications – Comparison of distillation and absorption Boiling point diagram and equilibrium diagram for a binary mixture - Raoult's law - Applications of Raoult's law Volatility - Relative volatility – Derivation of equation for relative volatility- Simple problems Ideal and non ideal solutions- Maximum and minimum boiling azeotrope Principles and applications of - Simple distillation - Equilibrium distillation - Steam distillation - Fractionation - Extractive and azeotropic distillation.			
CO2	**Interpret the types of distillation.**		
M2.01	Interpret simple distillation.	4	Applying
M2.02	Interpret equilibrium distillation.	3	Applying
M2.03	Interpret McCabe - Thiele method.	8	Applying
M2.04	Outline the constructional details of distillation equipment.	3	Understanding
M2.05	Identify the factors affecting efficiency of distillation column.	1	Understanding
	Series Test – I	1	

Contents:

Simple distillation - Derivation of Rayleigh's equation - simple problems

Equilibrium distillation - simple problems

Fractionation - Ideal plate - Material balance and heat balance equation for an ideal plate - McCabe & Thiele method with assumptions - Derivation for rectifying line and stripping line equations - 'q' factor - Values of 'q' based on five different feed conditions - McCabe and Thiele method for solving problems - Estimate the number of theoretical plates required for a given separation and locate the feed plate graphically – Limitations of McCabe Thiele method - Overall plate efficiency - Minimum reflux and total reflux - Pump around.

Sketch distillation column including condenser, reboiler, accumulator, trays, down comer, weir, distributor, chimney tray, feed line, distillate outlet, reflux and residue outlet - Functions of reboiler, condenser and accumulator in a distillation column - Outline distillation column internals including Distributor, Redistributor, Weir, Downcomer, Bubble cap tray, Sieve tray and Valve tray.

Efficiency of plate column – Entrainment- Weeping - Dumping – Flooding - Coning.

CO3	**Summarise leaching operation.**		
M3.01	Summarize the theory behind leaching operation.	4	Understanding
M3.02	Classify leaching operation.	2	Understanding
M3.03	Illustrate the working of leaching equipment.	7	Understanding
M3.04	Illustrate super critical fluid extraction.	2	Understanding

Contents:

Leaching - Principle - Applications - Factors affecting rate of leaching - Batch and continuous leaching operations

Classification of leaching operations.

Constructional details, diagram and working of Heap leaching, Shank system, Boll man extractor - Rotocel extractor, Kennedy extractor, Continuous Countercurrent Decantation (CCD).

Super critical fluid extraction – principle - applications.

CO4	**Summarise liquid – liquid extraction.**		
M 4.01	Summarize the theory behind liquid–liquid extraction	3	Understanding
M 4.02	Interpret ternary phase diagram.	3	Understanding
M 4.03	Differentiate distillation and extraction.	1	Understanding
M 4.04	Illustrate the working of extraction equipment.	4	Understanding
	Series Test – II	1	

Contents:
Liquid – liquid extraction - Principle – Applications – Differentiate solid –liquid extraction and liquid – liquid extraction – Distribution coefficient – Selectivity - Selection of solvent Classification of ternary systems – graphical representation of liquid – liquid equilibrium data (in equilateral triangular coordinates). Compare Distillation and Extraction. Constructional details and working of extraction equipment – Mixer settler - Packed column - Spray column - Rotating Disk contactor - Pulse column.

Text / Reference

T/R	Book Title/Author
T1	Mass Transfer Operations; Treybal.R.E
R1	Unit Operations. II; K A Gavhane -
R2	Unit Operations; McCabe and Smith
R3	Mass Transfer Operations; Surya Narayana
R4	Unit Operations of Chemical Engg. Vol. – I; P.Chathopadhya

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103145/